

1. SECTION: PRODUCT AND COMPANY IDENTIFICATION

Product name : 來莫畏
LIME-A-WAY

Other means of identification : Not applicable

Recommended use : Delimer

Product dilution information : 0.8 % - 100.0 %

Company : Ecolab Limited
10F., No.172, Sec.4, Chengde Rd., Shilin Dist., Taipei, Taiwan 111
(02) 8978-1234

Emergency telephone : +1-651-222-5352 (United States)

Issuing date : 10.11.2015

2. SECTION: HAZARDS IDENTIFICATION
GHS Classification
Product AS SOLD

Corrosive to Metals : Category 1
Skin corrosion/irritation : Category 1
Serious eye damage/eye irritation : Category 1

Product AT USE DILUTION

Corrosive to Metals : Category 1
Skin corrosion/irritation : Category 1
Serious eye damage/eye irritation : Category 1

GHS Label element
Product AS SOLD

Hazard pictograms :



Signal Word : Warning

Hazard Statements : May be corrosive to metals.
Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**
Keep only in original container. Wash skin thoroughly after handling.
Wear protective gloves/ protective clothing/ eye protection/ face protection. Do not mix with bleach or other chlorinated products – will cause chlorine gas. Do not breathe dusts or mists.
Response:
IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and

SAFETY DATA SHEET

LIME-A-WAY

easy to do. Continue rinsing. IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Take off contaminated clothing and wash before reuse. Absorb spillage to prevent material damage. Immediately call a POISON CENTER or doctor/ physician.

Storage:

Store in corrosive resistant/ .? container with a resistant inner liner.

Product AT USE DILUTION

Hazard pictograms

:



Signal Word

: Danger

Hazard Statements

: Causes severe skin burns and eye damage.
May be corrosive to metals.

Precautionary Statements

: **Prevention:**

Wash skin thoroughly after handling. Wear protective gloves/ protective clothing/ eye protection/ face protection. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Response:

IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Take off contaminated clothing and wash before reuse. Immediately call a POISON CENTER or doctor/ physician.

Storage:

Store in corrosive resistant/ .? container with a resistant inner liner.

Product AS SOLD

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Product AT USE DILUTION

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

3. SECTION: COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture

: Mixture

Chemical Name	CAS-No.	Concentration (%)
Phosphoric acid	7664-38-2	10 - 30
hydrochloric acid	7647-01-0	1 - 5

Product AT USE DILUTION

SAFETY DATA SHEET

LIME-A-WAY

4. SECTION: FIRST AID MEASURES

First aid measures for different exposure routes

Product AS SOLD

- | | | |
|---|---|--|
| In case of eye contact | : | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately. |
| In case of skin contact | : | Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately. |
| If swallowed | : | Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately. |
| If inhaled | : | Remove to fresh air. Treat symptomatically. Get medical attention. |
| Protection of first-aiders | : | If potential for exposure exists refer to Section 8 for specific personal protective equipment. |
| Notes to physician | : | Treat symptomatically. |
| Most important symptoms and effects, both acute and delayed | : | See Section 11 for more detailed information on health effects and symptoms. |

Product AT USE DILUTION

- | | | |
|-------------------------|---|--|
| In case of eye contact | : | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately. |
| In case of skin contact | : | Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately. |
| If swallowed | : | Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately. |
| If inhaled | : | Remove to fresh air. Treat symptomatically. Get medical attention. |

5. SECTION: FIRE-FIGHTING MEASURES

Product AS SOLD

- | | | |
|---------------------------------------|---|--|
| Suitable extinguishing media | : | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Unsuitable extinguishing media | : | None known. |
| Specific hazards during fire fighting | : | Exposure to decomposition products may be a hazard to health. |
| Hazardous combustion products | : | Decomposition products may include the following materials:
Oxides of phosphorus
Hydrogen chloride |

SAFETY DATA SHEET

LIME-A-WAY

Special protective equipment for fire-fighters : Use personal protective equipment.

Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

6. SECTION: ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

7. SECTION: HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers. To maintain product quality, do not store in heat or direct sunlight.

Storage temperature : 0 °C to 40 °C

SAFETY DATA SHEET

LIME-A-WAY

Product AT USE DILUTION

Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers. To maintain product quality, do not store in heat or direct sunlight.

8. SECTION: EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
Phosphoric acid	7664-38-2	TWA	1 mg/m ³	TW OEL
		STEL	2 mg/m ³	TW OEL
hydrochloric acid	7647-01-0	CEIL	5 ppm 7.5 mg/m ³	TW OEL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Safety glasses with side-shields
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Safety glasses with side-shields

SAFETY DATA SHEET

LIME-A-WAY

	Face-shield
Hand protection	: Wear the following personal protective equipment: Standard glove type. Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
Skin protection	: Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing
Respiratory protection	: When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

9. SECTION: PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Color	: clear, light green	green
Odor	: odorless	slight
pH	: 1.0, 100 %	1.0 - 2.2
Flash point	: Not applicable	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: No data available	
Flammability (solid, gas)	: No data available	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: 1.095	
Water solubility	: soluble	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	
Oxidizing properties	: No data available	
Molecular weight	: No data available	
VOC	: No data available	

10. SECTION: STABILITY AND REACTIVITY

SAFETY DATA SHEET

LIME-A-WAY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: Exposure to sunlight. Heat.
Incompatible materials	: Bases Metals Organic materials
Hazardous decomposition products	: No decomposition if stored and applied as directed.

11. SECTION: TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.
Ingestion	: Causes digestive tract burns.
Inhalation	: May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.
Ingestion	: Causes digestive tract burns.
Inhalation	: May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: Redness, Pain, Corrosion
-------------	----------------------------

SAFETY DATA SHEET

LIME-A-WAY

Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Toxicity

Product AS SOLD

Acute oral toxicity	: Acute toxicity estimate : > 5,000 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : 6.24 mg/l
Acute dermal toxicity	: Acute toxicity estimate : > 5,000 mg/kg
Skin corrosion/irritation	: No data available
Serious eye damage/eye irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

12. SECTION: ECOLOGICAL INFORMATION

Product AS SOLD

Ecotoxicity

Environmental Effects	: This product has no known ecotoxicological effects.
-----------------------	---

Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: No data available
Toxicity to algae	: No data available

Ingredients

Toxicity to daphnia and other aquatic invertebrates	: Phosphoric acid 48 h EC50 Daphnia magna (Water flea): > 100 mg/l
---	---

Ingredients

Toxicity to algae	: Phosphoric acid 72 h EC50 Desmodesmus subspicatus (green algae): > 100 mg/l
-------------------	--

Persistence and degradability

No data available

Bioaccumulative potential

SAFETY DATA SHEET

LIME-A-WAY

No data available

Mobility in soil

No data available

Other adverse effects

No data available

13. SECTION: DISPOSAL CONSIDERATIONS

Product AS SOLD

- Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

- Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

14. SECTION: TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

- UN number : 3264
- Description of the goods : CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.
(Phosphoric acid, Hydrochloric acid)
- Class : 8
- Packing group : II
- Environmentally hazardous : no

Sea transport (IMDG/IMO)

- UN number : 3264
- Description of the goods : CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.
(Phosphoric acid, Hydrochloric acid)
- Class : 8
- Packing group : II
- Marine pollutant : no
- Special precautions for user : None

Product AT USE DILUTION

Not intended for transport.

SAFETY DATA SHEET

LIME-A-WAY

15. SECTION: REGULATORY INFORMATION

National regulatory information

Regulations on Labelling and Hazard Communication of Hazardous Chemicals

National regulatory information

Regulations on Occupational Safety and Health Facilities

National regulatory information

Standards for the Storage, Cleanup, Handling and Disposal of Industrial Waste

National regulatory information

Rules on Road Traffic Safety

National regulatory information

Standard for the Control of Designated Hazardous and Dangerous Chemicals

National regulatory information

Standards of Permissible Exposure Limits in Workplace

The ingredients of this product are reported in the following inventories:

United States TSCA Inventory :

On TSCA Inventory

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL

Australia Inventory of Chemical Substances (AICS) :

On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemical Substances :

On the inventory, or in compliance with the inventory

Japan. ENCS - Existing and New Chemical Substances Inventory :

On the inventory, or in compliance with the inventory

Japan. ISHL - Inventory of Chemical Substances (METI) :

On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :

On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

On the inventory, or in compliance with the inventory

China. Inventory of Existing Chemical Substances in China (IECSC) :

On the inventory, or in compliance with the inventory

16. SECTION: OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet

Globally Harmonized System of Classification and Labeling of Chemicals (GHS)

IARC: (International Agency for Research on Cancer)

US. National Toxicology Program (NTP) Report on Carcinogens

SAFETY DATA SHEET

LIME-A-WAY

ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

Issuing date	: 10.11.2015
Version	: 1.5
Prepared by	: Regulatory Affairs
Local contact	: Ecolab Limited
Address	: 10F., No.172, Sec.4, Chengde Rd., Shilin Dist., Taipei, Taiwan 111
Telephone	: (02) 8978-1234
Name	: Regulatory Specialist
Position	: Regulatory Specialist

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.